# AI Agent Framework Comparison 2026

Comprehensive analysis of 9 open-source AI agent frameworks for self-hosted deployment

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## Executive Summary

This document evaluates 9 open-source AI agent frameworks across 6 weighted dimensions: Security (30%), Code Quality (20%), Orchestration (20%), Ecosystem (15%), Popularity (10%), and Hardware Efficiency (5%). The composite score reflects overall suitability for production self-hosted deployment.

## Comparison Table

| SCORE | AGENT | LANGUAGE | STARS | LICENSE | SECURITY | CODE QUALITY | ORCHESTRATION | ECOSYSTEM | POPULARITY | HARDWARE | MIN HW | DOCKER IMAGE | PORT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 7.5 | **IronClaw** | Rust | 4.8K+ | Apache 2.0 | 9/10 Very High | 9/10 Excellent | 7/10 Good | 6/10 | 4/10 Growing | 6/10 Light | 2C/4T, 2GB, 500MB | nearai/ironclaw | 18789 |
| 7.3 | **OpenHands** | Python | 65K+ | MIT | 7/10 High | 7/10 Good | 8/10 Excellent | 8/10 | 8/10 Very High | 3/10 Heavy | 4C/8T, 8GB, 10GB | ghcr.io/openhands/openhands | 3000 |
| 6.9 | **Moltis** | Rust | 2.3K+ | MIT | 7/10 High | 9/10 Excellent | 6/10 Fair | 7/10 | 3/10 Emerging | 8/10 Very Light | 1C/2T, 512MB, 100MB | moltis/moltis | 8080 |
| 6.3 | **NanoClaw** | Node.js | 21.5K+ | MIT | 7/10 High | 6/10 Fair | 7/10 Good | 5/10 | 6/10 Notable | 5/10 Moderate | 2C/4T, 4GB, 5GB | qwibitai/nanoclaw | Varies |
| 6.2 | **NemoClaw** | Node.js | 4.6K+ | NVIDIA | 9/10 Very High | 5/10 Fair | 6/10 Fair | 5/10 | 4/10 Growing | 3/10 Heavy | 4C/4T, 8GB, 5GB | nvidia/nemoclaw | 18789 |
| 6.1 | **Goose** | Rust + Python | 29K+ | Apache 2.0 | 5/10 Medium | 7/10 Good | 5/10 Fair | 9/10 | 7/10 High | 3/10 Heavy | 4C/8T, 8GB, 2GB | block/goose | 3000 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6.0 | **OpenClaw** | Node.js | 430K+ | MIT | 2/10 Low | 4/10 Poor | 9/10 Excellent | 10/10 | 10/10 Dominant | 5/10 Moderate | 2C/4T, 4GB, 2GB | alpine/openclaw | 18789 |
| 5.7 | **PicoClaw** | Go | 25K+ | MIT | 5/10 Medium | 6/10 Fair | 5/10 Fair | 5/10 | 7/10 High | 10/10 Minimal | 1C/1T, 10MB, 50MB | sipeed/picoclaw | 18789 |
| 5.4 | **NanoBot** | Python | 26.8K+ | Open Source | 5/10 Medium | 5/10 Fair | 6/10 Fair | 4/10 | 7/10 High | 8/10 Very Light | 1C/2T, 512MB, 200MB | hkuds/nanobot | 18790 |

## Agent Profiles

### IronClaw 7.5/10

**Language:** Rust **Stars:** 4.8K+ **License:** Apache 2.0 **Repo:** github.com/nearai/ironclaw

OpenClaw-inspired rewrite in Rust by Illia Polosukhin, co-author of the Transformer paper ("Attention Is All You Need") / NEAR AI. All untrusted tools run in WebAssembly sandboxes with capability-based permissions. Gateway API compatible with OpenClaw. Zero outbound API calls possible — fully air-gapped operation.

**Security:** Wasm isolation (seL4-inspired). Zero-default access. Endpoint allowlisting, credential injection, leak detection. FIPS certification path.

**Code Quality:** Pure Rust, zero unsafe code. Wasm formal verification. Capability-based security model throughout.

**Orchestration:** OpenClaw-compatible Gateway API, session management, MCP tools, capability-based task control.

**Ecosystem:** OpenClaw-compatible: MCP, multi-provider (Anthropic, OpenAI, Gemini, Mistral, Ollama), Wasm plugins, REST API.

**Pro:** Best security (Wasm sandbox, zero-trust), air-gap capable, Rust memory safety

**Con:** ~5% of OpenClaw skill library, 15ms overhead per skill, higher install friction

**Min Hardware:** 2 CPU cores, 4 threads, 2 GB RAM, 500 MB storage

### OpenHands 7.3/10

**Language:** Python **Stars:** 65K+ **License:** MIT **Repo:** github.com/OpenHands/OpenHands

AI-driven software development platform. Best-in-class REST API + WebSocket server. Docker-native with event-sourced state model. Raised $18.8M. Supports defining agents in code, running locally or scaling to 1000s in the cloud. Enterprise-ready with VPC deployment.

**Security:** Docker container isolation. Event-sourced state for audit/replay. Fine-grained access control.

**Code Quality:** Event-sourced, deterministic replay, SWE-bench tested. Python (no memory safety guarantees).

**Orchestration:** Full REST+WS API, Agent SDK for composing agents, multi-agent support, CI/CD integration, health endpoints, task abort.

**Ecosystem:** Enterprise: REST+WS API, Agent SDK, MCP, Docker runtime, event sourcing, K8s deploy.

**Pro:** Best REST API, Docker isolation, $18.8M funded, 87% bug tickets solved same day

**Con:** Resource-heavy (8GB+ RAM), looping on ambiguous tasks, high LLM API costs

**Min Hardware:** 4 CPU cores, 8 threads, 8 GB RAM, 10 GB storage

## Moltis 6.9/10

**Language:** Rust **Stars:** 2.3K+ **License:** MIT **Repo:** github.com/moltis-org/moltis

Single Rust binary — no Node.js, no npm, no runtime dependencies. Runs on Mac Mini, Raspberry Pi, or any server. Built-in voice, memory, scheduling, Telegram, Discord, browser automation, MCP servers. Streaming-first responses. Keys never leave your machine.

**Security:** Container sandbox. No telemetry, no phone-home. Full audit trail. Keys never leave machine.

**Code Quality:** 150K lines Rust, zero unsafe workspace-wide, 2,300+ tests, Prometheus/OTel, auditable agent loop (~5K lines).

**Orchestration:** JSON-RPC API, built-in sessions, memory system for context, cron scheduling, MCP tools, health endpoint.

**Ecosystem:** Built-in: MCP, voice, memory, scheduling, Telegram, Discord, browser, 8 TTS/7 STT providers.

**Pro:** Single binary, no runtime deps, voice+memory+MCP built-in, runs on Raspberry Pi

**Con:** Small community (2.3K stars), limited channel support, integration/tuning overhead

**Min Hardware:** 1 CPU core, 2 threads, 512 MB RAM, 100 MB storage

## NanoClaw 6.3/10

**Language:** Node.js (TypeScript) **Stars:** 21.5K+ **License:** MIT **Repo:** github.com/qwibitai/nanoclaw

Container-first agent built on Claude Agent SDK by Qwibit AI. Every agent session runs in an isolated Docker container. Supports "Agent Swarms" — teams of Claude instances collaborating in parallel. Docker partnership announced March 2026.

**Security:** OS-level container isolation per session. "Isolation over defense" — hard limits at container layer. Provable, auditable security.

**Code Quality:** 700 lines TypeScript. Tiny footprint. Container isolation compensates for minimal app-layer security. 62 contributors.

**Orchestration:** Container-per-session, Agent Swarms (parallel Claude instances), Docker Sandbox integration, multi-channel, container lifecycle.

**Ecosystem:** Focused: Claude SDK, Agent Swarms, Docker isolation, multi-channel, memory.

**Pro:** OS-level container isolation per session, Docker partnership, Agent Swarms

**Con:** Requires Anthropic API key (Claude only), heavier than lightweight alternatives

**Min Hardware:** 2 CPU cores, 4 threads, 4 GB RAM, 5 GB storage

## NemoClaw 6.2/10

**Language:** Node.js + OpenShell **Stars:** 4.6K+ **License:** NVIDIA **Repo:** github.com/NVIDIA/NemoClaw

NVIDIA enterprise security layer for OpenClaw. Announced at GTC 2026. Triple enforcement: Sandbox (Landlock + seccomp + network namespace), Policy Engine (filesystem/network/process rules), Privacy Router (inference routing + PII stripping). 85.6% PinchBench score.

**Security:** Landlock + seccomp + network NS. Policy engine + PII router. Enterprise-grade by NVIDIA.

**Code Quality:** Alpha stage. NVIDIA backing. "Expect rough edges." Not production-ready per NVIDIA's own documentation.

**Orchestration:** Via OpenClaw Gateway (inherits full API), policy-filtered context, OpenShell health monitoring, privacy router.

**Ecosystem:** Security stack: OpenShell sandbox, policy engine, privacy router, via OpenClaw ecosystem.

**Pro:** Triple enforcement (Landlock+seccomp+NS), PII router, NVIDIA backing, 85.6% PinchBench

**Con:** Alpha/not production-ready, Linux only, missing fleet management, limited model support

**Min Hardware:** 4 CPU cores, 4 threads, 8 GB RAM, 5 GB storage

## Goose 6.1/10

**Language:** Rust + Python **Stars:** 29K+ **License:** Apache 2.0 **Repo:** github.com/block/goose

By Block (Square/Cash App/Afterpay). Extensible AI agent with CLI and Electron desktop interfaces. Connects to 3,000+ tools through MCP protocol. Automates workflows with "recipes." Model-agnostic. 400+ contributors. 60% of Block workforce uses it weekly.

**Security:** No built-in sandbox. Red-teamed by Block security ("Operation Pale Fire"). Found and fixed prompt injection via invisible Unicode.

**Code Quality:** Rust core, red-teamed by Block security team, 400+ contributors, mixed Rust+Python codebase.

**Orchestration:** Extension protocol, CLI-focused (not API-first), recipes for workflows, MCP tools, no native sub-agents.

**Ecosystem:** Extensive: 3K+ MCP tools, recipes, extensions, Electron desktop, plugin system, grant program.

**Pro:** 3K+ MCP tools, Block backing, 50-75% dev time savings reported, free, model-agnostic

**Con:** No built-in sandbox, prompt injection found (red team), terminal-centric learning curve

**Min Hardware:** 4 CPU cores, 8 threads, 8 GB RAM, 2 GB storage

## OpenClaw 6.0/10

**Language:** Node.js **Stars:** 430K+ **License:** MIT **Repo:** github.com/openclaw/openclaw

The original open-source personal AI assistant. Fastest-growing open-source project in history (220K stars in 84 days). 2M monthly active users, 27M web visits/month. Built-in HTTP Gateway API. Connects to WhatsApp, Telegram, Discord, Slack. Multi-provider: Anthropic, OpenAI, Ollama.

**Security:** None built-in. No permission model, no code signing, full FS/shell/network access. 9 CVEs including CVSS 8.8 RCE (CVE-2026-25253). 341 malicious skills on ClawHub. 40K+ exposed instances.

**Code Quality:** 430K lines. Security score 28/100. 111 red flags. $373K technical debt. Poor test coverage (526h remediation needed).

**Orchestration:** Full REST+WS Gateway, session mgmt (thread modes), native sub-agents, agent teams with handoff protocols, cron, webhooks, task abort.

**Ecosystem:** Massive: 10K+ skills, ClawHub marketplace, MCP (3K+), webhooks, cron, 5+ channels, REST+WS API.

**Pro:** Largest ecosystem (10K+ skills), most integrations, 2M active users, battle-tested

**Con:** 9 CVEs incl. RCE, no sandbox, 341 malicious skills on ClawHub, credential leaks

**Min Hardware:** 2 CPU cores, 4 threads, 4 GB RAM, 2 GB storage

## PicoClaw 5.7/10

**Language:** Go **Stars:** 25K+ **License:** MIT **Repo:** github.com/sipeed/picoclaw

Ultra-lightweight Go agent by Sipeed. Under 10MB RAM, boots in 1 second. Runs on $5/$10 hardware. 99% less memory than OpenClaw. Rebuilt from the ground up in Go through a "self-bootstrapping" process. Native MCP integration, vision pipeline for multimodal LLMs.

**Security:** Cron security gating. Pre-v1.0 — "unresolved security issues." Not for production-critical workloads yet.

**Code Quality:** Go, lowest issue ratio (0.013), but 95% AI-generated code. Pre-v1.0. 187 contributors.

**Orchestration:** Gateway API (OpenClaw-like), basic sessions, spawn_status for sub-agents, cron, MCP.

**Ecosystem:** Growing: MCP, vision pipeline, sub-agents, cron, system tray UI, multimodal.

**Pro:** 10MB RAM, boots in 1s, runs on $5 hardware, single binary, vision pipeline

**Con:** Pre-v1.0 security caveats, limited complex workflows, basic state management

**Min Hardware:** 1 CPU core, 1 thread, 10 MB RAM, 50 MB storage

## NanoBot 5.4/10

**Language:** Python **Stars:** 26.8K+ **License:** Open Source **Repo:** github.com/HKUDS/nanobot

Ultra-lightweight AI assistant from HKU Data Intelligence Lab. ~4,000 lines of Python — 99% smaller than OpenClaw. Gateway mode with HTTP API. Supports subagent spawning for parallel tasks. "Auditable in an afternoon."

**Security:** Minimal sandboxing. Small codebase (4K lines) naturally limits attack surface. Relies on LLM provider security.

**Code Quality:** 4K lines Python, auditable in an afternoon, minimal tests, 215 contributors.

**Orchestration:** Gateway HTTP API, basic sessions, native subagent spawning, multi-provider, channel adapters.

**Ecosystem:** Minimal: Gateway API, subagent spawning, multi-provider, channel adapters.

**Pro:** Only 4K lines, easiest to understand/hack, multi-provider, subagent spawning

**Con:** Minimal sandbox, small ecosystem, limited advanced features compared to OpenClaw

**Min Hardware:** 1 CPU core, 2 threads, 512 MB RAM, 200 MB storage

## Score Methodology

| DIMENSION | WEIGHT | WHAT IT MEASURES |
|---|---|---|
| **Security** | 30% | Sandboxing approach, known CVEs, credential handling, audit results, zero-trust architecture |
| **Code Quality** | 20% | Language memory safety, test coverage, technical debt, code audits, unsafe code blocks |
| **Orchestration** | 20% | API programmability, session management, sub-agent support, task control, multi-agent coordination |
| **Ecosystem** | 15% | Tools, plugins, skills, MCP integrations, channels, protocols, marketplace |
| **Popularity** | 10% | GitHub stars, community size, media coverage, corporate backing, enterprise adoption |
| **Hardware** | 5% | Resource efficiency — inverse of minimum RAM/CPU requirements |

Security is weighted highest (30%) because agents handle sensitive data, credentials, and have system-level access. Popularity is weighted lowest (10%) because widespread adoption does not equal quality — OpenClaw is the most popular agent but has the worst security record.

## Sources

### Security Audits & Research

- Microsoft Security Blog: Running OpenClaw Safely — Identity, isolation, runtime risk analysis
- Cisco: Personal AI Agents Are a Security Nightmare — Enterprise security assessment
- Kaspersky: Key OpenClaw Risks — Credential exposure and infostealer targeting
- Block Red Team: Operation Pale Fire — Goose prompt injection via invisible Unicode
- Bitsight: OpenClaw Security Risks — 40,214 exposed instances, 35.4% vulnerable
- IronClaw Security Review — Wasm sandbox architecture and seL4-inspired capability model

### Code Quality & Architecture

- OpenClaw Audit: Modern Core with Hidden Debt — $373K tech debt, security 28/100, 111 red flags
- Moltis: Zero Unsafe, Full Audit Trail — 150K Rust lines, 2,300+ tests, zero unsafe
- Rust Agent Runtime Showdown — Moltis vs ZeroClaw architecture comparison
- OpenHands Software Agent SDK — Event-sourced architecture academic paper
- 116K Stars in 8 Weeks: Four Teams Rewrote OpenClaw — Codebase size and contributor comparisons

### Community & Popularity

- OpenClaw Statistics 2026 — 430K stars, 2M MAU, 27M web visits, 900+ contributors
- Goose AI Review 2026 — 29K stars, 400+ contributors, Block backing

- NanoClaw + Docker Partnership (TechCrunch) — Docker deal, community growth
- OpenHands Review 2026 — 65K stars, $18.8M funding, enterprise adoption
- PicoClaw GitHub — 25K stars, edge/IoT community

## Comparisons & Analysis

- Full 2026 Comparison: ZeroClaw vs OpenClaw vs NanoClaw vs NanoBot vs PicoClaw vs IronClaw
- OpenClaw Alternatives Compared (AI Magicx)
- NemoClaw at GTC 2026 (TechCrunch)
- Agent Wars 2026
- Security-First Guide to Self-Hosted AI Assistants

## Benchmarks

- SWE-bench Verified — Software engineering benchmark
- Terminal-Bench — Terminal agent benchmark
- PinchBench — OpenClaw agent performance benchmark

## Forums & Community Discussions

- Hacker News: NanoClaw + Docker Discussion
- Hacker News: Moltis Show HN Launch
- DEV Community: Claw Family Overview
- Product Hunt: IronClaw Launch